

## LIMOGES AP, France

WMO: 074340

Lat: 45.8611N

Lon: 1.1750E

Elev: 396

StdP: 96.66

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -5.3       | -3.6 | -10.0                         | 1.7 | -1.0 | -7.7 | 2.1 | -0.6 | 10.4                  | 6.4  | 9.5 | 6.3  | 3.5                   | 30   | 0.512 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 9.4                    | 30.6            | 20.1 | 28.5 | 19.5 | 26.8 | 18.6 | 21.2                | 28.4 | 20.3 | 26.9 | 19.4 | 25.4 | 3.1                  | 160  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 18.9                            | 14.4 | 24.2 | 18.0 | 13.6 | 23.1 | 17.1 | 12.8 | 22.1 | 63.2          | 28.4 | 60.0 | 26.9 | 56.9 | 25.5 | 27.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| (4) 8.7           | 7.7  | 6.9 | -7.2                       | 33.5 | 2.2                | 1.9 | -8.7                                               | 34.9 | -10.0      | 36.0 | -11.2      | 37.0 | -12.8      | 38.4 |  |  |
| (5)               |      |     | -7.8                       | 22.9 | 2.3                | 1.2 | -9.4                                               | 23.8 | -10.8      | 24.5 | -12.1      | 25.1 | -13.7      | 26.0 |  |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 11.6   | 4.4   | 4.9   | 8.1   | 10.4  | 13.8  | 17.4  | 19.2  | 19.3  | 15.9  | 12.8  | 7.6   | 5.2  |
|                                                                                  | DBStd       | 6.53   | 3.66  | 4.08  | 3.75  | 3.74  | 3.78  | 3.94  | 3.60  | 3.67  | 3.38  | 3.70  | 3.64  | 3.84 |
|                                                                                  | HDD10.0     | 709    | 174   | 147   | 83    | 39    | 8     | 1     | 0     | 0     | 1     | 16    | 89    | 152  |
|                                                                                  | HDD18.3     | 2633   | 431   | 377   | 318   | 240   | 148   | 64    | 33    | 31    | 88    | 173   | 323   | 408  |
|                                                                                  | CDD10.0     | 1302   | 1     | 3     | 23    | 51    | 126   | 222   | 286   | 289   | 179   | 103   | 16    | 2    |
|                                                                                  | CDD18.3     | 183    | 0     | 0     | 0     | 1     | 8     | 36    | 60    | 62    | 16    | 2     | 0     | 0    |
|                                                                                  | CDH23.3     | 1484   | 0     | 0     | 0     | 8     | 58    | 300   | 492   | 517   | 106   | 5     | 0     | 0    |
|                                                                                  | CDH26.7     | 424    | 0     | 0     | 0     | 0     | 5     | 80    | 143   | 178   | 18    | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 3.4    | 3.7   | 3.8   | 3.9   | 3.7   | 3.4   | 3.2   | 3.1   | 2.8   | 3.1   | 3.3   | 3.5   | 3.7  |
| Precipitation                                                                    | PrecAvg     | 1001   | 91    | 80    | 81    | 85    | 94    | 73    | 62    | 74    | 77    | 85    | 96    | 99   |
|                                                                                  | PrecMax     | 1447   | 196   | 221   | 222   | 187   | 198   | 161   | 166   | 208   | 222   | 161   | 226   | 249  |
|                                                                                  | PrecMin     | 729    | 25    | 6     | 4     | 6     | 20    | 0     | 4     | 10    | 7     | 6     | 19    | 2    |
|                                                                                  | PrecStd     | 142    | 46    | 45    | 49    | 49    | 39    | 39    | 36    | 42    | 49    | 43    | 46    | 54   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 13.3  | 17.2  | 20.9  | 24.5  | 27.3  | 32.1  | 33.0  | 34.2  | 28.9  | 23.9  | 18.1  | 14.1 |
|                                                                                  |             | MCWB   | 9.0   | 9.6   | 11.7  | 14.5  | 17.9  | 21.0  | 20.3  | 20.6  | 18.1  | 16.9  | 13.1  | 10.3 |
|                                                                                  | 2%          | DB     | 11.5  | 13.9  | 18.2  | 21.4  | 24.9  | 28.8  | 30.1  | 30.8  | 26.2  | 21.6  | 15.7  | 12.4 |
|                                                                                  |             | MCWB   | 9.3   | 8.5   | 10.8  | 13.4  | 16.3  | 19.9  | 20.1  | 19.9  | 17.2  | 16.1  | 12.0  | 9.8  |
|                                                                                  | 5%          | DB     | 10.4  | 11.7  | 16.0  | 19.2  | 22.8  | 26.8  | 28.0  | 28.4  | 24.1  | 19.9  | 14.1  | 11.2 |
|                                                                                  |             | MCWB   | 8.7   | 8.1   | 10.0  | 12.2  | 15.7  | 18.8  | 19.4  | 19.5  | 16.8  | 15.4  | 11.6  | 9.4  |
|                                                                                  | 10%         | DB     | 9.2   | 10.0  | 13.8  | 16.9  | 20.6  | 24.7  | 26.1  | 26.1  | 22.0  | 18.1  | 12.6  | 10.1 |
|                                                                                  |             | MCWB   | 7.7   | 7.7   | 9.0   | 11.1  | 14.7  | 18.0  | 18.5  | 18.6  | 16.0  | 14.6  | 10.8  | 8.6  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 11.2  | 11.0  | 12.8  | 15.6  | 19.2  | 22.5  | 22.1  | 22.1  | 20.2  | 18.2  | 14.2  | 11.7 |
|                                                                                  |             | MCDB   | 11.7  | 14.4  | 18.0  | 22.6  | 25.4  | 28.8  | 29.2  | 29.6  | 25.2  | 21.8  | 16.3  | 12.8 |
|                                                                                  | 2%          | WB     | 10.0  | 9.7   | 11.6  | 14.2  | 17.6  | 20.8  | 21.0  | 21.1  | 18.8  | 16.9  | 13.0  | 10.6 |
|                                                                                  |             | MCDB   | 10.9  | 12.1  | 16.3  | 20.3  | 23.3  | 27.1  | 28.2  | 28.7  | 24.0  | 20.1  | 14.8  | 11.7 |
|                                                                                  | 5%          | WB     | 9.0   | 8.9   | 10.6  | 12.8  | 16.4  | 19.6  | 20.0  | 20.2  | 17.6  | 16.0  | 12.0  | 9.8  |
|                                                                                  |             | MCDB   | 10.0  | 10.7  | 14.5  | 17.8  | 21.3  | 25.5  | 26.6  | 26.9  | 22.3  | 18.8  | 13.6  | 10.9 |
|                                                                                  | 10%         | WB     | 8.0   | 8.0   | 9.7   | 11.6  | 15.1  | 18.4  | 19.1  | 19.2  | 16.6  | 15.1  | 10.9  | 8.9  |
|                                                                                  |             | MCDB   | 9.0   | 9.6   | 12.8  | 15.9  | 19.6  | 23.7  | 25.1  | 25.1  | 20.8  | 17.7  | 12.5  | 9.8  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 5.1   | 6.3   | 7.7   | 8.3   | 8.8   | 9.2   | 9.4   | 9.4   | 8.8   | 7.3   | 5.7   | 5.2  |
|                                                                                  |             | MCDBR  | 6.1   | 9.2   | 10.8  | 11.6  | 11.8  | 11.9  | 12.0  | 12.4  | 11.7  | 9.2   | 7.5   | 6.4  |
|                                                                                  |             | MCWBR  | 4.5   | 5.7   | 5.9   | 6.1   | 6.1   | 5.9   | 5.4   | 5.2   | 5.5   | 5.2   | 4.9   | 4.6  |
|                                                                                  | 5% WB       | MCDBR  | 4.9   | 7.0   | 9.2   | 10.4  | 10.6  | 11.1  | 11.2  | 11.1  | 10.1  | 7.9   | 6.6   | 5.2  |
|                                                                                  |             | MCWBR  | 4.1   | 4.9   | 5.5   | 5.7   | 6.2   | 5.8   | 5.6   | 5.2   | 5.4   | 4.8   | 4.7   | 4.2  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.303  | 0.319 | 0.348 | 0.378 | 0.380 | 0.393 | 0.384 | 0.379 | 0.349 | 0.345 | 0.319 | 0.300 |      |
|                                                                                  | taud        | 2.494  | 2.445 | 2.375 | 2.289 | 2.329 | 2.326 | 2.346 | 2.373 | 2.436 | 2.459 | 2.487 | 2.492 |      |
|                                                                                  | Ebn at Noon | 809    | 861   | 879   | 879   | 887   | 874   | 877   | 868   | 868   | 817   | 778   | 770   |      |
|                                                                                  | Edh at Noon | 67     | 84    | 104   | 124   | 123   | 124   | 120   | 112   | 97    | 82    | 67    | 60    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.21   | 2.10  | 3.26  | 4.46  | 5.21  | 6.02  | 5.88  | 5.20  | 4.11  | 2.56  | 1.44  | 1.09  |      |
|                                                                                  | RadStd      | 0.15   | 0.27  | 0.41  | 0.57  | 0.47  | 0.56  | 0.41  | 0.43  | 0.27  | 0.21  | 0.18  | 0.17  |      |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only            |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (22 neighbors) |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page